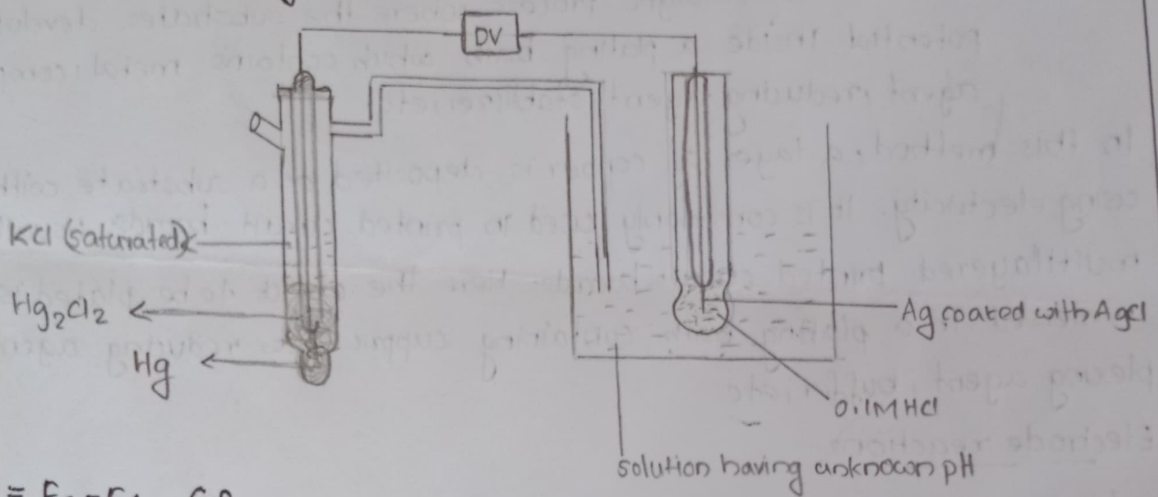


11. (a) How is glass electrode used in determining the pH of a solution? What are the advantages and limitations of glass electrode?
- (b) Describe the principle and applications of Electroless copper plating.

ans: (a) To determine pH, glass electrode is immersed in the unknown solution and coupled with SCE. The Na^+ ions in the outer layer is exchanged with the H^+ ions. so at equilibrium, a potential difference is developed between two surfaces (due to difference in conc. of H^+ ions). The potential difference developed is proportional to the pH of two solutions. The emf is then measured using a potentiometer or digital voltmeter.



$$E_{\text{cell}} = E_C - E_A = (E_G^\circ - 0.0591 \text{ pH}) - 0.2422 = (E_G^\circ - 0.2422) - 0.0591 \text{ pH}$$

This equation is in the form $Y = c + mx$. It is a straight line equation.

Intercept, $c = E_G^\circ - 0.2422$; Slope, $m = -0.0591$ at 25°C (298K)

When a graph is plotted between emf along Y-axis and pH along X-axis. The constant c and m can be determined by calibrating the pH meter with two buffer solutions of known pH. Value of m at any other temperature is obtained on calibration. The pH of the test solution can be calculated by noting the emf of the cell with test solution and knowing the values of c and m .

Advantages

→ Simple to operate.

- Used to determine the pH of oxidising, reducing and even coloured solution.
- It is not easily poisoned.
- Used to determine pH between 0 and 12.

Limitations

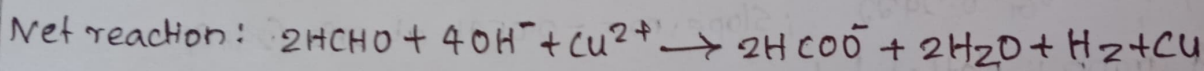
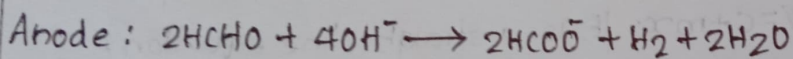
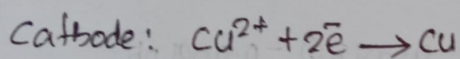
- Glass material possesses high resistance. So ordinary potentiometer cannot be used.
- It cannot be used for determining pH above 12.
- It cannot be used in solutions of acetic acid, ethyl alcohol and gelatin.

b) Electroless copper plating

Principle: It is an autocatalytic process where the substrate develops a potential inside a plating bath which contains metal, complexing agent, reducing agent, stabilizer, etc.

In this method, a layer of copper is deposited on a substrate without using electricity. It is commonly used in printed circuit boards. Usually, multilayered printed circuit boards. Here the object to be plated is immersed in a plating bath containing cupric salt, reducing agent, complexing agent, buffer, etc.

Electrode reactions



Applications

- widely used for metalizing printed circuit boards.
- Used for plating on non-conductors.
- It is also used for making decorative plating on plastics.

12 a. Write the principle and procedure for the estimation of ferrous ion using dichromate solution potentiometrically. $E^{\circ} \text{Fe}^{3+}/\text{Fe}^{2+} = 0.77 \text{V}$

ans: Principle

Potential of an electrode depends on the concentration of ion. A change in ionic concentration changes the electrode potential. During the titration, concentration of ion changes and as a result potential of indicator electrode changes which in turn changes the emf of the cell.

eg: Oxidation of Fe^{2+} to Fe^{3+} using Ce^{4+} in acid
Reduction potential of $\text{Ce}^{4+}/\text{Ce}^{3+} \rightarrow 1.6 \text{V}$
 $\text{Fe}^{3+}/\text{Fe}^{2+} \rightarrow 0.77 \text{V}$

Procedure

20 ml Ferrous Sulphate $[\text{FeSO}_4]$ solution is taken in a beaker. 20 ml dil Sulphuric acid is added to it to make the medium acidic.

A platinum electrode is inserted and coupled with saturated calomel electrode.

Potential difference is then measured using a digital voltmeter or potentiometer.

Standard Ce^{4+} solution is taken

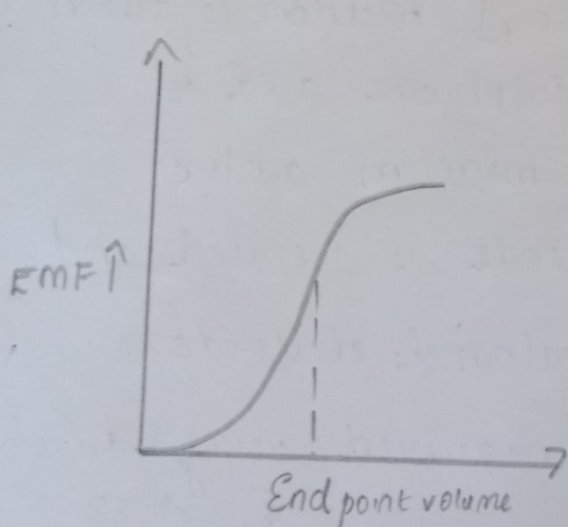
in a burette and added to acidified FeSO_4 in 1 ml installments. EMF is then noted after each addition of Ce^{4+} . Then EMF is plotted against volume of Ce^{4+} .

When Ce^{4+} is added, concentration of Fe^{3+} increases due to oxidation of Fe^{2+} . So potential of indicator electrode increases.

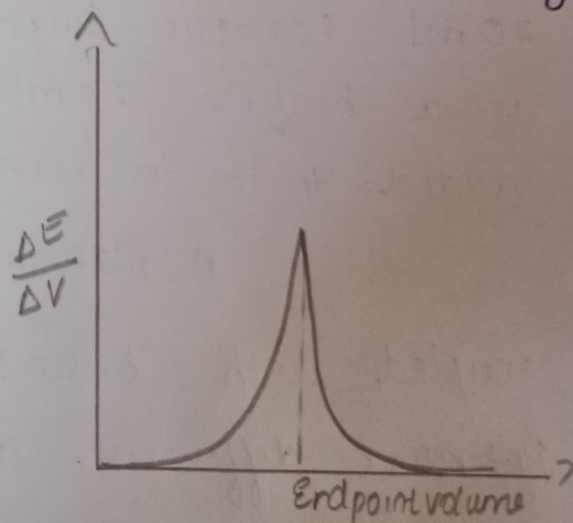
$$E_{\text{indicator}} = E^\circ + \frac{0.0591}{n} \log \frac{[\text{Fe}^{3+}]}{[\text{Fe}^{2+}]}$$

At the endpoint, no more Fe^{2+} was for oxidation. So $\text{Ce}^{4+}/\text{Ce}^{3+}$ electrode begins to setup. So potential increases sharply. At the end point, rate of change of EMF, i.e. $\frac{\Delta E}{\Delta V}$ is max.

Before the endpoint, potential is due to $\text{Fe}^{3+}/\text{Fe}^{2+}$ (0.74) system. After the endpoint $\text{Ce}^{4+}/\text{Ce}^{3+}$ electrode setup (1.4). So potential increases sharply.



Vol of $\text{Ce}^{4+} \rightarrow$



Vol of Ce^{4+}

12b. Emf of an electrochemical cell is 1.5175V at 20°C and 1.5213 V at 35°C. If the cell reaction involve 2 electrons, find the standard emf of the cell and the reaction quotient

ans. $E_{\text{cell}} = 1.5175\text{V}$ at 20°C & 1.5213 at 35°C

$$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{2.303 RT}{nF} \log Q$$

A plot of EMF against temperature must be a straight line having intercept = E°_{cell}

$$\text{slope} = - \frac{2.303 R}{nF} \log Q$$

$$E = A + BT$$

$$1.5175 = A + 293 B$$

$$1.5213 = A + 308 B$$

$$\text{Solving } -3.8 \times 10^{-3} = -15B$$

$$\text{ie } B = 2.53 \times 10^{-4}$$

$$\therefore A = 1.5175 - 293 B$$

$$= 1.5175 - 293 \times 2.53 \times 10^{-4} = 1.443$$

$$E^{\circ}_{\text{cell}} = 1.443\text{V} = \text{standard emf}$$

$$\therefore B = - \frac{2.303 R}{nF} \log Q$$

$$2.53 \times 10^{-4} = \frac{-2.303 \times 8.314}{2 \times 96500} \log q$$

$$\log q = \frac{2.53 \times 10^{-4} \times 2 \times 96500}{-2.303 \times 8.314}$$

$$= \frac{48.829}{-19.147}$$

$$\log q = -2.553$$

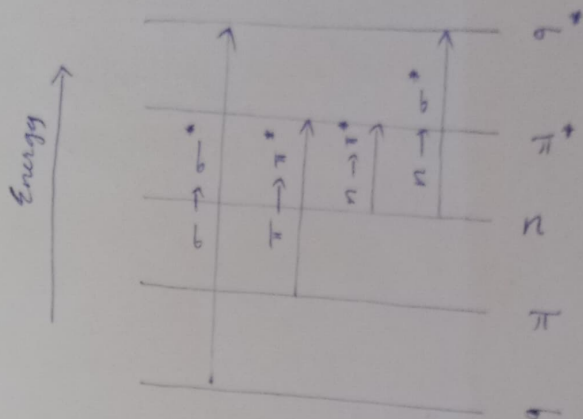
$$\text{ie } q = 2.79 \times 10^{-3}$$

Ques 13

a) What are the various types of electronic transitions possible in organic molecule? Give examples of each. Also give the instrumentation of UV visible Spectrophotometer.

The possible electronic transitions in organic molecules are:

- ① $\sigma \rightarrow \sigma^*$
- ② $\pi \rightarrow \pi^*$
- ③ $n \rightarrow \pi^*$
- ④ $n \rightarrow \sigma^*$

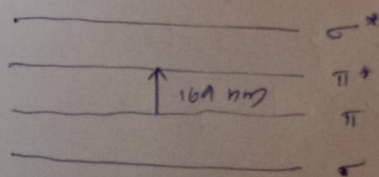


1. $\sigma \rightarrow \sigma^*$ transition

Saturated hydrocarbons show $\sigma \rightarrow \sigma^*$ transition of C-C σ bond and C-H σ bond. This type of transition occurs below 150 nm. Ordinary UV machines can take the spectrum only from 200 - 700 nm. So saturated hydrocarbons cannot be detected using ordinary UV machines. Here the transition energy required is very high.

2. $\pi \rightarrow \pi^*$ transition

Unsaturated hydrocarbons containing double or triple bonds show this type of transition. Eg: C=C in ethylene molecule. It gives an absorption maxima at 164 nm. Here the transition energy required is comparatively low. Molecular energy level diagram of ethylene:



This transition can be observed in conjugated dienes.

3. $n \rightarrow \pi^*$ transition

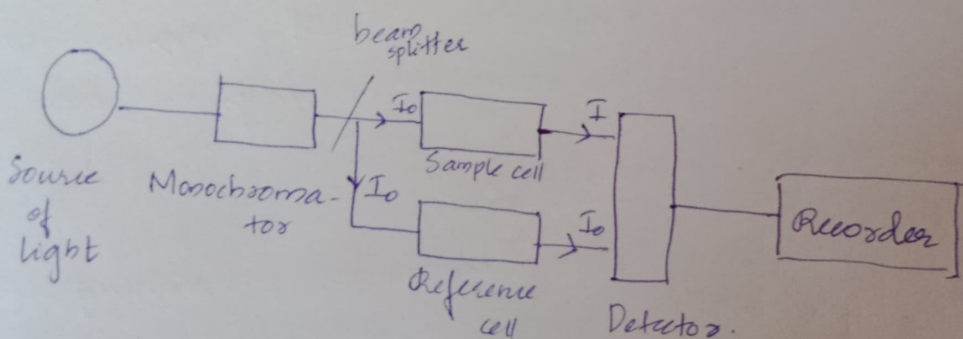
Such transitions are observed for aldehydes and ketones which contain carbonyl (C=O) group. Carbonyl group has both π electrons and non bonding electrons. This transition will give a less intense spectrum compared to $\sigma \rightarrow \sigma^*$ and $\pi \rightarrow \pi^*$, since it is a symmetry forbidden transition.

eg: Formaldehyde, conjugated unsaturated aliphatic ketones.

4. $n \rightarrow \sigma^*$ transition

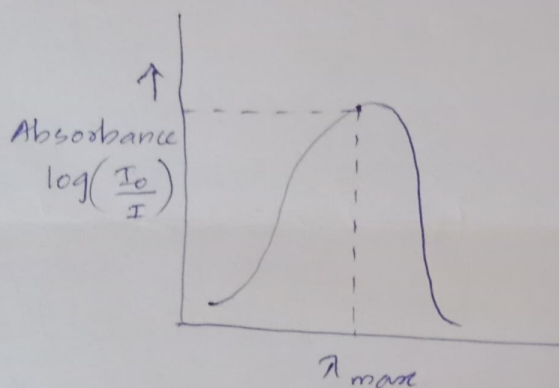
Compound containing S, N, O etc which contain non bonding electrons show this type of transition. It is also symmetry forbidden transition. Absorption take place below 200 nm.

INSTRUMENTATION OF UV-VISIBLE SPECTROPHOTOMETER

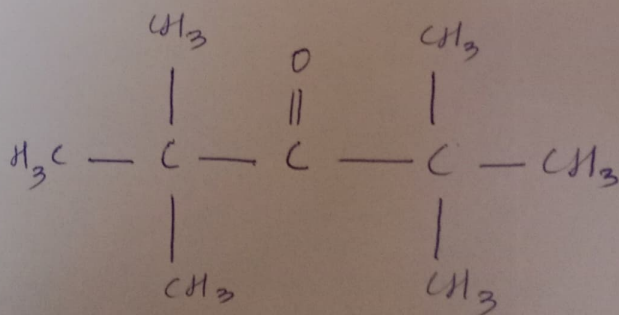
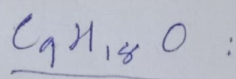
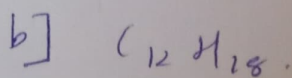
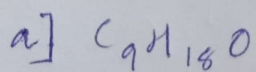


Radiation from the source (xenon flash lamp or hydrogen discharge lamp) is passed through a monochromator. Monochromator allows a narrow range of wavelength pass through it. The beam is coming out of the monochromator is split into

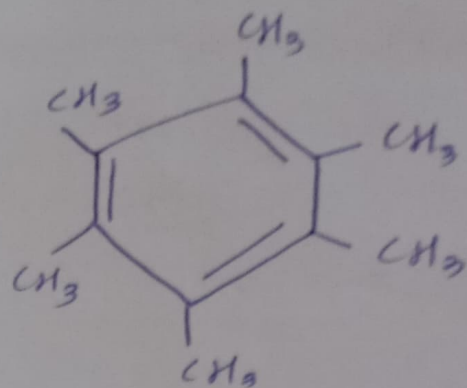
two equal beams. having equal intensities. One of the beam is passed through the sample and the other one through the reference. Intensities of these beams are measured by the detector and then compared. The detector converts radiation into current. The current $\propto$ concentration of the solution. The signal from the detector is finally, received by the recorder which gives the output in the form of a graph. in which absorbance is plotted against wavelength. The wavelength of maximum absorbance is a characteristic value designated as λ_{max} and different compounds have different λ_{max} .



b) Suggest structural formula for the following compounds such that they give a single signal in proton NMR spectroscopy.



C₁₂H₁₈



Write the various types of vibration possible for HCl , CO_2 and H_2O and state which of these modes are IR active. Write reason for their IR activity

For a molecule to be IR active, there should be a change in dipole during the vibrational motion of molecule

• HCl

It is a heteronuclear diatomic molecule.

$\therefore$ No. of atoms, $n = 2$

$$\begin{aligned}\text{It is linear, so number of vibrational modes} &= 3n - 5 \\ &= 3 \times 2 - 5 \\ &= 6 - 5 \\ &= 1\end{aligned}$$

Therefore, HCl exhibits stretching vibration and is IR active because it is hetero nuclear.

• CO_2

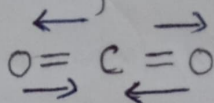
It is a triatomic linear molecule

$\therefore$ No. of atoms, $n = 3$

$$\begin{aligned}\text{No. of vibrational modes} &= 3n - 5 \\ &= 3 \times 3 - 5 \\ &= 9 - 5 \\ &= 4\end{aligned}$$

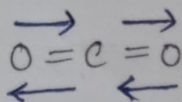
The possible vibrations are

(i) Symmetric stretching



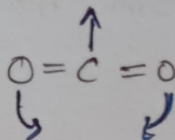
As there is no change in dipole moment, it is IR inactive.

(ii) Asymmetric stretching

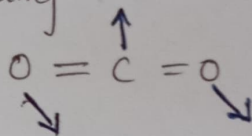


It is IR active as change in dipole is observed.

(iii) In plane Bending



(iv) Out of plane Bending



Both in plane and out of plane bendings are IR active as there is change in dipole moment.

• H_2O

It is a non-linear triatomic molecule

$\therefore$ No. of atoms, $n = 3$

No of vibrational modes $= 3n - 6$

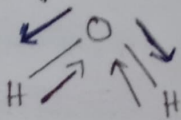
$$= 3 \times 3 - 6$$

$$= 9 - 6$$

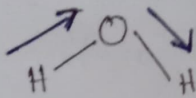
$$= 3$$

It exhibits

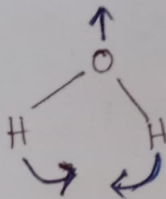
(i) Symmetric stretching



(ii) Asymmetric stretching

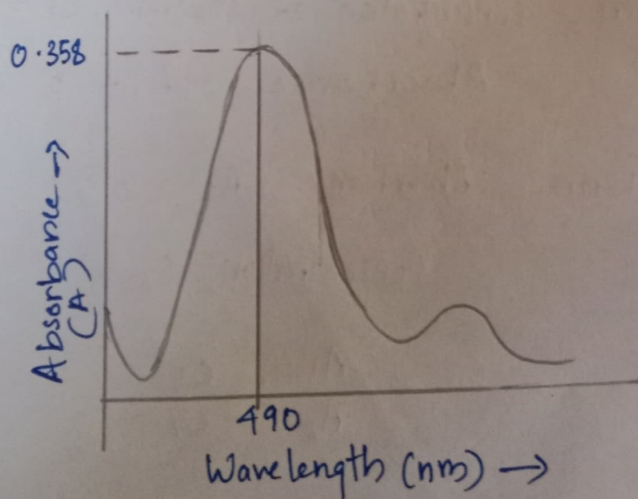


(iii) Inplane bending



In all the above cases, there is difference in dipole moment, so they are all IR active.

(b) State Beer-Lambert's law and explain the term molar extinction co-efficient. Given is the absorption spectrum of a compound A of 2.5×10^{-6} M concentration, when measured using 1cm cuvette in a UV-Vis spectrometer. Calculate the unknown concentration of a test sample of compound A if the absorbance is 0.518, when measured in the same condition.



When a parallel beam of monochromatic electromagnetic radiation is passed through a homogeneous absorbing medium, the rate of decrease in intensity of radiation with thickness of the medium is directly proportional to the intensity of incident radiation at that point and concentration of the solution. This is known as Beer Lambert's Law.

$$\log \frac{I_0}{I} \propto ct$$

$$\log \frac{I_0}{I} = \epsilon_f \cdot c \cdot t$$

I_0 - intensity of incident radiation

I - intensity of transmitted radiation

c - concentration of solution

t - thickness of medium.

ϵ_f - Molar absorptivity or molar extinction coefficient.

It depends on the nature of medium and also the extent of absorption.

We know that $\log \frac{I_0}{I} = \text{Absorbance (A)}$

$$A = \epsilon_f ct$$

$$A \propto c$$

Given, concentration of solution, $c_1 = 2.5 \times 10^{-6} \text{ M}$

Absorbance, $A_1 = 0.358$ (from graph)

When, Absorbance, $A_2 = 0.518$

Concentration, $c_2 = ?$

$$\frac{A_1}{A_2} = \frac{c_1}{c_2}$$

$$\frac{0.358}{0.518} = \frac{2.5 \times 10^{-6}}{C_2}$$

$$C_2 = \frac{2.5 \times 10^{-6} \times 0.518}{0.358}$$

$$= 3.62 \times 10^{-6} \text{ M}$$